

The Magic2 deep RNA-sequencing analyser

User Guide

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Abstract

Magic2 is an accurate and sensitive long and short reads RNA deep-sequencing analyser. It can read local fasta/fastq files or import data-sets on the fly from the NCBI short-read-archive (SRA). It can export in a single pass the alignment files, the coverage plots, the gene expression and many metrics. It is very easy to use because it analyzes in a dry run the top of the sequence files and the hardware to select the optimal strategy. This guide explains how to use this program.

Magic2 is freely available from [git@github.com:NLM-DIR/Magic2](https://github.com/NLM-DIR/Magic2)

Contents

1 Introduction

Considering the fast evolution of the sequencing technologies, we need to accurately aligned an ever greater number of short paired-end Illumina reads and ever longer reads from PacBio, Nanopore, Roche and other manufacturers. The files are very large and it is important to minimize the number of steps and intermediate formats.

Magic2 can align local fasta/fastq files but it can also automatically download SRR runs from NCBI short read archives (SRA). Optionally, these data can be cached on the local computer as new fastq files, but if you have a fast network connection to NCBI, this is not mandatory. Conversely, Magic2 is convenient just to download the data that you wish to analyze with other programs.

The program recognizes the eventual adaptors and reports very detailed quality control metrics, It is fast because, in a single pass, it can export the alignments, the gene expression counts, the known and new introns and their support, the polyA addition sites. and a collection of coverage plots indicating the gene expression, the probable breakpoints and the candidate transcriptions starts and stops.

Section 1 gives simple examples of how to use Magic2. Section 2 explains how to download and recompile the program. Section 3 gives all the details on the recognized sequence formats and the way to download the runs from SRA. Section 4 explains the construction of the target index. The remaining section present the various output files and how to use them.

If available, the code will delegate parts of the calculations to a GPU.

2 Simple examples

First download the code on a Linux or Mac machine with at least 32 Giga bases or RAM. The code will already run if you have 4 processors, and will run faster on a larger machine because it automatically adapts its level of paralellism to the host computer.

2.1 Align data on a raw genome

To just align a set of reads just on a genome, you need to type two successive commands

```
1 magic2 --createIndex IDX.test -t genome.fasta.gz
2 magic2 -x IDX.test -i read.fasta.gz -o out
```

where 'genome.fasta.gz' is a single fasta file containing the sequence of all the chromosomes and 'read.fasta.gz' contains the reads. The results will be visible in the output directory 'out'. The data files can be gzip compressed or not, the sequences can be given in fasta or fastq format.

To align paired end reads, the pairs can be given in two fasta/fastq files separated by a plus sign, or in interleaved format

```
1 magic2 -x IDX.test -i read.R1.fastq.gz+read.R2.fastq.gz -o out2a
2 magic2 -x IDX.test -i read.sample_12.fastq.gz -o out2b
```

Finally, the data can be downloaded on the fly from the NCBI Short-Read-Archives (SRA) and analyzed simply by providing a set of SRR identifiers

```
1 magic2 -x IDX.test -srr SRR1234,SRR1235,SRR1236 -o out23
```

Notice that you not need to say if the sequences are DNA or RNA, long or short, or if they contain adaptors. All these properties are automatically recognized by the code by performing a dry run on the first 10 Mega bases, the realigning everything with the propoer strategy.

2.2 Align data on an annotated genome

More accurate and more detailed result will be obtained if you provide a known annotation. Magic2 will favaozize jumping the known introns, and will report the gene expression counts. It will also report the fraction of the sequences mapping on the ribosomal RNA (sometimes as high as 80%) and on the mitochondria. This is important because a complete genome, like the human T2T project, contains very many repetitions of the ribosomal RNAs and echos of parts of the mitochondria, and many programs, including Magic2, would not report highly repetitive alignments.

To provide an annotation, you need to create a small target configuration file called for example T2T.tConfig containing 2 tab separated columns

```
1 G      T2T.chromosomes.fasta.gz
2 M      T2T.mitochondria.fasta.gz
3 R      T2T.rRNA.fasta.gz
4 I      T2T.gtf.gz
```

where T2T.gff.gz is the T2T annotation file, available for example from NCBI or EBI in gtf or gff format.

Then, create an annotated index by using the parameter (-T) rather then (-t), and align

```
1 magic2 --createIndex IDX.T2T -T T2T.tConfig
2
3 magic2 -x IDX.T2T -srr SRR1234,SRR1235,SRR1236 --full -o out5
```

If you have a very large machine, you could process the 3 runs in parallel, the memory needed by the index will be shared, but still you will consume larger ressources

```
1 magic2 -x IDX.T2T -srr SRR1234 --full -o out5a &
2 magic2 -x IDX.T2T -srr SRR1234 --full -o out5b &
3 magic2 -x IDX.T2T -srr SRR1236 --full -o out5c &
```

In the ouput directory, you will find the alignemnt files, the polyAs, the intron counts and if you used the -full parameter, yo will also find the gene expression counts the canditate transcription starts and ends and a collection of coverage plots.

3 Downloading and installing Magic2

3.1 Simply download the executables

The precompiled executable files can be found at

```
1 where are the binaries
```

select your machine (Mac or Linux) and eventually also download the GPU enabled code

3.2 Download the source code and recompile

The code is open-source and freely available from github.

```
1 git clone git@github.com:NLM-DIR/Magic2
```

In case of very intensive use, the precompiled executables may be slightly slower than the programs recompiled locally because the latter may be better tailored to the instruction sets of your actual hardware. To recompile the C code, stay in the download directory and type

```
1 tcsh
2 cd ./Magic2
3 setenv ACEDB_MACHINE LINUX_4_OPT
4 make libs
5 cd wsa
6 make
```

The executable will be the file `./Magic2/bin.LINUX_4_OPT/magic2`

// //

In this project, our main goal is completeness and accuracy. We hope to align the highest possible percentage of reads and of bases in each read, even in the presence of sequencing noise. To simulate a high noise level, we aligned primate sequences, from Chimpanzee to Marmoset against the T2T human genome. As a truth control for accurate splicing, we aligned the full length RefSeq mRNAs and a contrario several DNA sample which a priori should not skip introns.

Our secondary goal is to simplify the user's experience. The program does not require any cost or strategy parameters. It evaluates by itself, via a dry run on the first 10 mega bases, the sequencing technology and the characteristic of the sample and automatically recognizes and clips the eventual adapters. It also evaluates the hardware to decide the optimal level of parallelisation and if available will run part of the calculations on a GPU.

Thanks to its non-standard parallelization by GO like channels, computing the coverage plots and the gene expression counts come at essentially no cost, whereas in current methods (HISAT, Star etc.) they usually require a very slow sorting and rescanning of the BAM files. Actually, in Magic2, exporting the BAM alignment files can be considered optional, since Magic2 may be able to export directly all the data useful to your project.

A collateral benefit of Magic2 is its speed. The computers, over the last ten to twenty years, have considerably accelerated the computing units (CPUs), but memory access kept lagging. To compensate, multi-layered cache architectures were designed. The CPUs are grouped in cores, and access the large memory (the RAM in giga bytes) via several level of caches (typically L1 in kb, L2 in Mb). Considering that memory access is the bottle neck, we designed the Magic2 algorithms to privilege sequential memory access. For example

4 Presentation of the Query language

4.1 Overview

The first aim of our query language is to allow the systematic exploration of the database by the construction of large tables that can join the data contained in families of objects, subject to textual and arithmetic filtering. In other words, the query language allows to give a relational view of our object oriented database. The syntax of a complex query is reminiscent of SQL, it takes the form

```
1  select x, z from x in ..., y in ..., z in ... where .... order ...
```

In SQL, the variables would iterate through tables. In an object oriented database, the first variable x typically iterates across all the members of a given class, then y is derived *from* x , then z is derived *from* x and/or y . The resulting (x, y, z) tuples are filtered by the *where* condition. Finally, only the (x, z) columns specified in the *select* clause, with lines sorted according to the *order* clause, are exported. The resulting syntax is very natural because in our normal life, we are more accustomed to reason about objects than about relations. For example, the meaning of the rather complex query

```
1  select A, P                                // select 2 variables: A and P
2    from A in class Person,                  // A scans the whole class Person
3    P in A->papers                            // P scans the papers of A
4    where count { A->papers } >= 5           // limit to at least 5 papers
```

should not be too difficult to grasp. In this first query, the variable A iterates across all members of class *Person*, the variable P through all the papers of each author and finally only the authors having at least 5 papers are selected.

Calculations can be performed on numerical values:

```
1  select p, bmi                               // Report persons with high body mass index
2    from p in class Person,                   // p scans all persons
3    h in p->height,                           // h is the height in meters
4    w in p->weight,                           // w is the weight in kilograms
5    bmi = w / h^2                             // bodymass index in kg/m^2
6    where bmi > 25                           // select overweight persons
```

In this second query, using the classical definition of the bodymass index, we select people who are heavy relative to their height. The intermediate variables h and w are not exported, but if they are not available in the database, the bmi will not be computed. By combining more variables, and using arithmetics and `{}` protected embedded subqueries, as in these examples, one can construct large tables which remain relatively easy to develop and maintain. The details on the exploration of the database are explained in section 2, the *where* clause in sections 3, the dates, DNA and protein formats in sections 4, ordering in section 5.

The second aim of the query language is to allow to express very simple things in a very simple way, as in the following examples

```

1 Find author      // list all members of class author
2 select K*        // limit to names starting with K
3 follow papers    // list their papers
4 select ?Person k* ; >papers    // idem on a single line
5 select ?sequence kinesin* ; DNA    // export a set of DNA sequences

```

The simplifications (detailed in section 2.5) are obtained in two ways. On the one hand, the queries are implicitly chained, either by being issued in succession, or by being separated by a semi-column. This allow to write only very short query pieces and try them one after the other. On the other hand, the computer is in charge of interpolating the syntactic sugar. For example, the computer expands the second line *select K** into *select p from p in @ where p like 'K*'*, where @ (the active set) refers to the results of the previous query. The short form contains all the information but is simpler, easier to grasp, easier to type, and less prone to clerical mistakes.

The details of the implementatin of the query language within the acedb system are described in the annex. The lists of all the keywords is presented in A.1. The history of the development is summarized in A.2. The way to download, compile and test the latest version of acedb is explained in A.3. A complete interactive query session is detailed in A.4. Finally annex A.5 presents the C-language API used in client programs.

4.2 Introduction to the acedb schema

In acedb, each object belongs to a class, with a known schema defined in the file `wspec/models.wrm`. A simple schema for a bibliography database can be

```

1 ?Paper Title UNIQUE ?Text      // Class Text is fast searchable
2   Author ?Person XREF Papers   // by default, tags are multivalued
3   Journal UNIQUE Text          // except if UNIQUE is specified
4   Submitted UNIQUE DateType    // Dates have a special format
5   Published UNIQUE DateType    //
6   Pages Int UNIQUE Int         // several linked values are allowed
7
8 ?Person Papers ?Paper XREF Author // XREF maintains the reverse relation
9   Affiliation Text             // Multivalued by default

```

A few instances of these classes can be

```

1 Person Tom
2   Professor
3   Papers p1 // Tom has published several papers
4             p2
5   Affiliation Paris
6             Tokyo
7
8 Person Jim
9   Papers p1
10  Affiliation London
11
12 Paper p1
13   Title "Quantum computers"
14   Author Tom
15           Jim
16   Journal "Nature"
17   Published 2001–04
18   Submitted 2000–11
19   Pages 1    23
20         100 107
21
22 Paper p2
23   Author Jim
24   Published 2004–06
25   Pages 17 19

```

A simple biometric and filiation schema can be specified as

```

1 ?Person Filiation Parent ?Person XREF Child
2                               Brother ?Person XREF Brother
3                               Child   ?Person XREF Parent
4           Biometrics   Height UNIQUE Float // Height in meters
5                               Weight UNIQUE Float // Weight in kilograms

```

The cross referencings (XREF) in the schema guarantee that the author/paper, the parent/child and the brother/brother relations remain synchronized.

4.3 Navigating through the acedb schema

As explained previously, one can list all existing papers using

```

1 select ?Paper

```

resulting in a 1 column table

```
1    p1
2    p2
```

One can also directly access a single object using

```
1    select OBJECT( 'Paper' , 'p1' )
```

Each object in the database has a name, belongs to a class and is time-stamped to the last time a value was modified. These fields can be accessed as methods

```
1    select p in ?Paper , p.name , p.class , p.timestamp
```

We can get to a tag using #, and get the value behind a tag using ->. For example the following query

```
1    select p in ?Paper , a in p->Author , prof in a#Professor , z in a->Affiliation
```

will produce the 4 column table

```
1    p1  Jim      NULL      London
2    p1  Tom  Professor  Paris
3    p1  Tom  Professor  Tokyo
4    p2  Jim      NULL      London
```

Reporting the pages creates a new challenge. According to the schema, the pages tag supports a two columns table of integers. To access the two columns, one must use square brackets.

```
1    select P in ?Paper , p1 in P->pages , p2 in p1[1] where p2
```

or the equivalent query, in which p sits on the pages tag of the object

```
1    select P,p1,p2 from P in ?Paper ,p in P#pages ,p1 in p[1] ,p2 in p[2] where p2
```

or the equivalent query, in which the bracket is associated directly to the tag name

```
1    select P,p1,p2 from P in ?Paper ,p1 in P->pages[1] ,p2 in P->Pages[2] where p2
```

In each case, one obtains

```
1    p1 1    23
2    p1 100 107
3    p2 17   19
```

Several details in the second form of the query should be noticed. a) The variable names are case sensitive, *p* and *P* are distinct. b) In principle *p*[2] should iterate through each number in column 2 behind *p*

```
1    select P, 'last page' ,p2 from P in ?Paper ,p in P#pages ,p2 in p[2] where p2
```

gives

1	p1	last	page	23
2	p1	last	page	107
3	p2	last	page	19

but when we have 2 bracketings of the same variable, they remain synchronized so $p[1]$ and $p[2]$ always refer to the same line of the table behind the tag *pages*. This synchronization is very important when a tag is multivalued. Consider a schema where each chromosome contains the position of all its genes

1				
2	Chromosome	chr_18		
3	Gene A	1000	2000	
4	Gene B	4000	6000	

The 3 equivalent queries

1	select	c,g,a1,a2	from	c	in	?chromosome,g	in	c->Gene,a1	in	g[1],a2	in	g[2]	
2	select	c,g,a1,a2	from	c	in	?chromosome,g	in	c->Gene,a1	in	g[1],a2	in	a1[1]	
3	select	c	in	?chromosome,	g	in	c->Gene,	a1	in	g[1],	a2	in	a1[1]

gives the expected answer, with one line per gene, with the correct coordinates

1	chr_18	A	1000	2000
2	chr_18	B	4000	6000

because on each line the coordinates $a1$ and $a2$ are derived from, and therefore are associated to, the gene g . But if we write the following more complex query where we reinitialize the definition of the 2 coordinates $a1, a2$ at the root of the object

1	select	c,	g,	a1,	a2					
2		from	c	in	?chromosome,					
3		g	in	c->Gene,	a1	in	c->Gene[2],	a2	in	c->Gene[3]

$a1, a2$ are no longer associated to the variable g and we obtain the silly combinatorial answer

1	chr_18	A	1000	2000
2	chr_18	A	1000	6000
3	chr_18	A	4000	2000
4	chr_18	A	4000	6000
5	chr_18	B	1000	2000
6	chr_18	B	1000	6000
7	chr_18	B	4000	2000
8	chr_18	B	4000	6000

4.4 COUNT, MIN, MAX, SUM, AVERAGE, STDDEV

The MIN, MAX, SUM, AVERAGE or STDDEV of any numerical variable, for example a multi-valued numerical field, can be evaluated by interpolating in capitals the desired operator in front of its definition

```
1  select p,x from p in ?Paper , x in MAX p->pages
2  select cc,x from cc in ?chromosome, x in AVERAGE cc->gene[2]
```

Rather than giving in x the list of all values associated to the query, one per line, one obtains for each object the required result as a single value.

4.5 Multi-valued data cells and transitive closure

In an object oriented database, tags are often multivalued. As already explained, the expand operator `->` gives access to the values behind a tag, one value per line. But in a table with several columns, if several tags are developed, this leads to a combinatorial number of lines. Suppose *tag1* has 4 values and *tag2* has 6 values, the query `select @->tag1, @->tag2` would give 24 lines. To limit this proliferation, it is sometimes more convenient to export all the values in a single data cell. The parallel-expand operator `=>` provides this possibility.

```
1      // => gives the parallel expansion, unsorted
2      // The results appear in the same order as in the object
3      // as desired when listing the authors of a paper
4  select ?paper p1; =>author
5      Tom; Jim // all authors come on a single line
6
7      // Genealogy example
8  select m,'is the mother of', m=>child from m in ?Person where m == eva
9      Eva is the mother of Cain; Abel; Seth
```

Another modification is the transitive-expand operator `>>` which gives the transitive closure of a tag, meaning that the `>>` operator is used iteratively. For example *grand_ma -> child* lists all the direct children of *grand_ma*, but *grand_ma >> child* gives her children, grand-children, great-grand-children ... For example, if the first two biblical generations are known in the database

```
1  select p, 'is a parent of', from p in ?Person, c in p->child where c
2      Enosh is a parent of Baraki
3      Enosh is a parent of Keman
4      Eva  is a parent of Abel
5      Eva  is a parent of Cain
6      Eva  is a parent of Seth
7      Seth is a parent of Enosh
```

The whole progeny of Eva is given by the *child* transitive closure

```

1   select ?Person Eva ; @ >> child // whole progeny
2   Abel
3   Baraki
4   Cain
5   Enosh
6   Keman
7   Seth

```

4.6 Simplification rules

The most frequent way to define the first variable is to iterate through a class. To list all members of the class sequence known in the database, one may write

```

1   select x from x in class "sequence" // full syntax, or equivalently
2   select x in class "sequence" // drop from since x is exported
3   select x in class sequence // drop the quotes
4   select class sequence // drop the name of the variable
5   select ?sequence // use ? as a shortcut for class
6   find sequence // intuitive equivalent syntax

```

The successive short cuts are all acceptable, because the original syntactic sugar was not bringing any information. First, the purpose of the *select ... from ...* syntax is to export only some of the arguments that are needed to compute, or to order, the exported columns in a specific way. If all columns should be exported the '.... *from* ' clause is no longer needed. Then, the double quotes around the class names are optional, as class names in acedb never contain spaces or non alphanumeric characters. The variable name *x* is redundant since *x* was not reused. The very frequent keyword *class* can be abbreviated as a leading question mark. Finally *find*, at the beginning of a query, is used as an alias of *select class*. The general idea is to allow the most concise syntax and remap it into a full query which is executed by the database.

Suppose we want to find all the members of *class Person* called *King*. Again this name does not contain any blank space, so starting from the full syntax, we can write any of the following equivalent queries

```

1   select x from x in class "Person" where x like "king" // full syntax
2   select ?Person like "king" // drop x and the where keyword

```

When the variable is not named, everything to the right of the class name is interpreted as being part of the *where* clause. One can replace the *like* by *~* (tilde), or replace it by one of the usual comparators *<* *<=* *==* *>=* *>*

```

1   select ?Person < kj // select Abel to King, but not Kong

```

Finally, if there is nothing else in the query we can eliminate the operator symbol

```

1      select ?Person King      // which is interpreted as
2      select a in ?Person where ( a#King or a like "King" )

```

we select at the same time the persons named 'King' or containing the tag *King*, In most situations, this is not ambiguous but using this simplified syntax to select a person with the meaningful name King may also select the current *King* of the kingdom !

The double quotes are not always needed. Indeed the comparison operators expect a declared variable, a number or a string on the right hand side. We do need to protect strings containing spaces or special characters, like slash or minus. We must also protect strings matching the reserved word, i.e. "select", "from" or strings matching the name of the declared variables, or tag names like in the previous example distinguishing the name "*king*" and the tag *King*. Attention, variable names are case sensitive, *x* and *X* are distinct, but reserved words (*select*, *Select*, *SELECT*) are not. Consider these 2 examples

```

1      select x in ?Person , y in x->Brother where y > x      // x is a variable
2      select x in ?Person , y in x->Brother where y > "x"    // "x" is letter x

```

The first query will select pairs of persons (*x*,*y*) where the name of *brother y* if alphabetically behind the name of *brother x*, for example (Abel, Cain), but not (Cain, Abel). Whereas the second query requests that *y* starts with x, y or z and would accept (Zachary, Xavier), but reject (Abel, Seth).

There is a special issue concerning the naming of the instances of a class. By default the instance names are not case sensitive, but the first typography encountered when parsing the data is always preserved. For example if in the datafiles the first spelling is "*Eva*", then the query *Find person eva* will return *Eva*. However, one may specify in *acedb* that the instance names of a particular class are case sensitive, this is specified in the self documented configuration file *wspec/options.wrm*. This option is used for example in the class *Gene* for the model organism *Drosophila melanogaster* which, according to tradition, has two distinct genes, one called *A* and the other called *a*.

Finally, on the command line, the leading keyword *select* can be abbreviated as *s*,

```

1      select ?Person k*      // full form
2      s ?Person k*          // s is a short form for select

```

All these shortcuts allow to type quickly terse queries on the command line interface, which are first mapped in a well defined way into a full fledged query language, then evaluated.

4.7 Chaining queries using the active list

The second most frequent way to derive the first variable of a query is to iterate through the active list. This concept was inspired by the Unix pipe. The active list is called @. It is a mathematical

set, always sorted, and with no duplicates. It starts empty. It is then populated or modified by each successive *select* statement. This allows to chain the queries like in:

```

1      select a from a in class "Person"           // populate the active list @
2      select a from a in @ where a like "king"      // derive a from the list
3
4      // which can be simplified into:
5      find person
6      select king

```

If a query exports several columns, the active list corresponds to the first exported column. For example

```

1      select p, j from p in class paper, j in p->journal // case PJ
2      select ?paper, >journal // equivalent short form of case PJ
3      select j, p from p in class paper, j in p->journal // case JP
4      select j      from p in class paper, j in p->journal // case J

```

Case PJ will export a list of (paper, journal) tuples and the active list contains papers. Whereas case JP will export (journal, paper) tuples and the active list will contain journals, and case J will only export journals and the active list will also contain journals.

4.8 Silent queries

In many situations, one may want to run a query to know the number of objects satisfying the query without wishing to see the resulting table. As in mathematica and other interactive languages, the output can be suppressed simply by adding a semi-column at the end of the query.

```

1      select ?Person ; // silent query endding with semi-column

```

Capitalizing on the concept of an active list and on the syntax shortcuts explained in the two previous sections, the semi-column can be used to chain queries:

```

1      select ?Person ; // all persons, silent
2      select Tom ;     // restrict to Tom, silent
3      select >papers   // export papers of author Tom

```

The result of the first query, in this case the set of all persons, is not exported, but is used as input to the second query, which selects author Tom, then to the third query. In fine this chained query exports the list of papers authored by Tom.

The same query can be presented on a single line:

```

1      select ?Person ; tom ; >papers // export Tom's papers

```

5 Filtering on arithmetic and textual conditions

5.1 Comparators

The *where* clause is used to filter the results. The simplest form is to search for the specific name of an object

```
1 select ?Person Tim
```

which expands as

```
1 select p in class "Person" where p == "Tim" // select a single person
```

The usual comparators: `<` `<=` `==` `>=` `>` can act on names, dates and numbers. In addition, a numeric variable is interpreted as a real number and numerical comparisons can involve arithmetics: additions, subtractions, multiplications, divisions, modulo, power. Parenthesis are recognized in the usual way:

```
1 select p in ?Person ,
2     h in p->height ,           // assume h is given in meters
3     w in p->weight             // assume w is given in kilograms
4     where 2 * (h - 100) < 1.7 * w // select relatively heavy people
```

In text comparisons, the value of a variable is its name, with the caveat that `acedb` interprets embedded numbers as numbers, implying the ordering

```
1 a < a7b < a11c < b
```

String matching comes in 2 flavors. Using the syntax *a* like *b*, or equivalently *a* ~ *b*, one can invoke a simple system with 2 kinds of jokers: question mark (?) to represent a single character or star (*) to represent an arbitrary string.

```
1 p ~ "T?m" // selects Tam, Tim and Tom,
2           // but not Attim which does not start with T
3 p ~ "T*m" // also selects Theotym,
4           // but not Thomas which does not end with m
5 p ~ "*T?m*" // also selects Attim"
6 p ~ "*T*m*" // selects Tam, Tim, Tom, Theotym, Thomas and Attim
```

Single quotes prevents the expansion of the (*) and (?) symbols

```
1 p == '*' // select person actually named *
```

5.2 Regular expressions

Alternatively, one may write *a* equal-tilde *b* to invoke the full C regular expression matching

```

1      p =~ 't[io]m'      // selects Tim, Tom and Attim but not Tam
2      p =~ '^t[io]m'     // selects Tim and Tom,
3                               // but not Attim which does not start with T
4      p =~ 't.*m'        // selects Tam, Tim, Tom, Theotym, Thomas and Attim
5      p =~ 't.*m$'       // rejects Thomas which does not ends with m

```

The regular expressions must be protected by single quotes to prevent premature evaluation.

5.3 Automatic classification using dynamic subclasses

In most situations, the class of a variable is known from the schema. Requesting *paper*→*author* would be known in our examples as yielding an author, i.e. an instance of *class Person*. However, acedb allows dynamic classification into sub-classes. For example, one can define in acedb a subclass *Prolific_author* as all authors of at least 5 papers known in the database. Whenever an *Person* exceeds this threshold, it belongs at the same time to the *class Person* and to the subclass *Prolific_author*. To get the full list of prolific authors one may directly list the subclass. But it may be necessary to check the status of a given set of authors, this is done using the operator *ISA* (is an instance of a class) as follows

```

1      // if Prolific_author is a subclass of class Person one may
2      select PA in class Prolific_author      // list the subclass
3                               // or select objects belonging to a subclass
4      select A ...      // select persons to populate the @ list
5      select PA from PA in @ where PA ISA Prolific_author // filter

```

The subclasses are defined in acedb in the configuration file *wspec/subclasses.wrm*

5.4 Combining Boolean conditions: AND, OR, XOR and NOT

As in nearly all query systems, conditions can be combined using (*OR*, *XOR*, *AND*, *NOT*) in that order of precedence and parentheses. The names of the operators are not case sensitive: (*XOR*, *Xor*, *xor*) are equivalent. (*OR*, *XOR*, *AND*, *NOT*) can be abbreviated as (*||* *^^* *&&* *!*). Remember that the single caret is reserved for arithmetic power $2^3 = 8$. For example, one may write

```

1      ... where (a <= x && x <= b) || (b < x && x < a)

```

to specify that *x* is either in the segment *[a,b]* or in the segment *]b,a[*. However, as discussed below, missing data and multivalued variables slightly complicate the semantics of the Boolean operators.

5.5 Dealing with missing data: TRUE, FALSE, NULL and NaN

In a standard programming language, like C, it is standard practice to assume that the numerical value zero matches the Boolean value FALSE. For example a looping instruction *while(x)* is equivalent to *while(x != 0)*. A standard error, however is to evaluate a non initialized value, but this

can be detected at compile time and fixed before running the program. Apart from zero, non-zero and non initialized values, a 4th type of number occurs in a program, the infamous NaN (Not A Number), which is generated as the results of an invalid arithmetic operation like $1/0$ or $\log(0)$. Unfortunately, these are only detected at run time, and may occur very frequently for example in *Deep Learning* programs (a branch of artificial intelligence).

In a database query, it is the rule, rather than the exception, to stumble upon a non initialized value. Indeed, all fields start empty. For this reason, the logical tests in the query language are not binary but ternary, the 3 allowed values being TRUE, FALSE and NULL.

Boolean tags, $p\#tag$ evaluate as NULL if p is not defined, otherwise as TRUE if the object p contains the tag , or FALSE if it does not.

Values, e.g. $(p \rightarrow weight)$, evaluate as NULL if undefined, i.e. if the object p is not defined, or if the tag $weight$ is not present in the object, or if the tag does not point to a value in the object. Otherwise they evaluate as TRUE even if the value happens to be zero.

In numerical calculations, the NaN value is sticky. Any calculation involving one NaN evaluates to NaN, for example $\log(0) + 4$. Similarly, any calculation involving a missing data evaluates to NULL, for example $(p \rightarrow weight + p \rightarrow height)$ evaluate as NULL if either $weight$ or $height$ is not specified in the object p .

Any string comparison involving a NULL value (missing data) and any number comparison involving a NaN or a NULL evaluates to FALSE.

Count operations always evaluate to a valid number.

In Boolean operations (AND, OR, XOR, NOT), NULL and NaN evaluate as FALSE.

This may seem a little abstract, but is easy to understand on examples, and important because, in a database query, undocumented values are very frequent.

1	select A from A in ?Person where A	// Always true
2	select A from A in ?Person where A#Professor	// Tom is a Professor
3	select A from A in ?Person where A→Professor	// False, no value
4	select P from P in ?Person where P→weight	// Is there a value
5	select P from P in ?Person where P→weight > 80	// Check the value
6	select 1,x from x = (-1)^0.5 where x	// False: not a number
7	select 1,x from x = (-1)^0.5 where 1 OR x	// True: T or NaN = T

Combining the filters, one can for example quality check the data and export all cases where the values are either missing or out of range, for example a weight should never be negative:

1	select p,w from p in ?Person, w in p→weight where w <= 0 OR NOT w
---	---

5.6 The meaning of NOT in a multivalued universe

In standard arithmetics, the two conditions $(a \neq b)$ and $!(a == b)$, i.e. (a NOT EQUAL b) versus (NOT (a EQUAL b)), are equivalent, they are FALSE if $(a == b)$ and TRUE otherwise. But in the context of a database query, the situation is more involved. In general, the variables are

multivalued, and the meaning of NOT depends on a choice of strategy. Consider the case of two papers, *p1* with authors *Tom* and *Jim*, and *p2* just with author *Tom*. We certainly expect Paper *p1* to answer TRUE to the query (*author == Tom*). Therefore it should answer FALSE to the query (*!(Author == Tom)*). But what about (*Author != Tom*). Scanning through the author list of *p1* we get 2 answers {TRUE, FALSE} and we need a strategy to reduce this list to a single answer. We chose to favor TRUE, i.e. the OR value of all individual answers. Therefore *p1* (but not *p2*) answers TRUE to (*Author != Tom*) because *p1* has at least one other author.

In conclusion, if the variables are multi-valued, the following queries are not equivalent

```

1  select p from p in class Paper where p->author != Tom
2  select p from p in class Paper where NOT p->author == Tom

```

With our choice of strategy: TRUE wins over FALSE. So in the first case, one find all papers where Tom is not the only author in the second case one finds all papers where Tom is not an author,. In addition, according to the previous section discussing missing data, the papers without a known list of authors are also selected in the second query, but not in the first query.

5.7 Counting elements in embedded subqueries

A very useful constraint is counting. For example, prolific authors can be found by either by counting the number of values of a tag, or the number of lines of an embedded subquery delimited by curly brackets {}, i.e. they can be found by the 2 equivalent queries

```

1  select a in ?Person where count a->papers > 5
2  select a in class author where count {select p in a->papers} >= 5

```

but in the latter case, we can become very specific and only count papers published in a given journal

```

1  select a in class "Person" where
2  count {select p in a->papers where p->journal == "nature"} >= 2

```

6 Basic data types

6.1 Constants and numerical calculations

A constant is declared using the equal symbol. The constant may be a number or a string. These constants can then be used in calculations and filters

```

1  select x, y, z from x = 2 , y = "hello", z = 2 * 3 + 4
2  select L from L in class line , pi = 3.14 where L->length > 2 * pi / 3
3
4  select d from d = 2 where d == 5 - 3    // correct filter
5  select d from d = 2 where d = 5 - 3    // error, please use ==

```

As usual in C and many other languages, setting the value of a constant uses the = sign, but assessing an equality in a *where* clause uses the == symbol. Looking closely, we may notice that since all the variables *x*, *y*, *z*, *L*, *d* are exported, the *select - from* is redundant, moreover the *x* variable is never reused, so its declaration is redundant. This lead us to the simplified forms:

```

1  select x from x = 2 // full form
2  select x = 2        // short form
3  select 2            // shortest form

```

Since the system understands the parenthesis and the standard arithmetic operators (plus, minus, multiply, divide, power, modulo) we actually have a (multi dimensional) calculator

```

1  select (1 + 2) * (3 - ( 4 + 5)) // returns (3) * (-6) = -18
2  select 8 modulo 3              // returns 2
3  select -1 modulo 3             // returns 2 (math convention)
4                                // rather than -1
5  select 2 * 5 modulo 3          // returns 1
6  select 9 ^ 2                   // returns square(9) = 9 * 9 = 81
7  select 9 ^ (1/2)               // returns sqrt(9) = 3
8  select (-1) ^ .5               // sqrt(-1)returns NULL, arithmetic exception
9  select 3*2 , (5+1)/2           // returns 6,3

```

When computing *-1 modulo 3*, we prefer the mathematical convention 2 because the C language convention *-1* breaks the periodicity of the modulo function and introduces complications when translating DNA codon triplets. All numerical calculations are performed using 64-bits floating numbers (C doubles).

6.2 Loose and strict date comparisons, date differences

Dates are special, because they may be specified with a very variable precision, sometimes just a year is provided, sometimes down to the second. To allow more meaningful comparisons we introduce two kinds of comparators, using either the lower or the higher precision.

A constant date is declared by enclosing the numbers using the back-quote sign:

```

1  select d = '2016-01-30_22:47:15 '

```

The successive numbers represent Y:year, M:month (1 to 12), D:day (1 to 31), h:hour (0 to 24), n:minute (0 to 60) and s:seconds (0 to 60).

In a user provided date, the leading zeroes may be dropped, they are reinstated in the answer:

```

1  query:: select d = '2016-1-17_3:7 ' // No leading zeroes
2  answer:: 2016-01-17_03:07           // standardized date format
3  query:: select d = '20160321'      // No minus signs
4  answer:: 2016-03-21                // standardized date format

```

In a database, the objects are usually dated with a certain granularity. For example the submission date of papers often gives the day, but for books, we often only know the year. Also a frequent query would be to retrieve the papers of a given year or a given month. To satisfy these needs, we introduce 2 kinds of date comparisons, strict and loose. In a strict comparison, using the comparators `<=` `==` `>=` the 2 dates are compared at the highest available precision, just like one would compare real numbers. In the loose mode, using the tilde comparators `<~`, `~`, `=~`, `>~`, the 2 dates are compared at the lowest precision, if one date is given in years, the other is rounded in years. As a result, we have

```

1      '2016-2' == '2016'           // —> false
2      '2016-2' >= '2016'          // —> true
3      '2016-2' <= '2016'          // —> false
4
5      '2016-2' =~ '2016'           // —> true
6      '2016-2' >~ '2016'          // —> true
7      '2016-2' <~ '2016'          // —> true
8
9      '2016-2' == '2016-2-17'      // —> false
10     '2016-2' == '2016-3'          // —> false
11
12     '2016-2' =~ '2016-2-17'      // —> true
13     '2016-2' =~ '2016-3'          // —> false

```

Notice that a date must always be protected by a pair of back-quotes. This need is obvious if you want to specify the month, since the date `'2016-3'`, meaning march 2016, is not the same thing as the subtraction $2016 - 3 = 2013$, but it is also needed even when you just give a year like `'2013'` to trigger the interpretation of 2013 as a date.

Finally, like in SQL, it is possible to compute a date difference at a desired precision using `DATEDIFF (unit, date_1, date_2)` where unit can be one of (year, month, day, hour, minute, second), or abbreviated as (y,m,d,h,n,s), notice the n for miNute. The result is an integer number.

In all cases, be very careful when using dates and always check the validity of the query on a few known cases since it is very easy, when dates are involved, to mean something and write something else.

6.3 DNA and proteins, motif searches

A specificity of `acedb` is that it understands the genetic code. If the database contains sequences, or mRNAs, or any object with a DNA tag, one can obtain its DNA using

```

1      select s in class sequence, d in DNA(s)
2      // yields the full DNA of each sequence in the database.
3      // To obtain a specific part use
4      select x = 3, y = 8, s in ?sequence, d in DNA(s,x,y)

```

```

5 // The reserved word DNA must be written in upper case, this
6 // allows to use dna as a variable name or to access
7 // the DNA tag. If x > y, one obtains the reverse complement
8 select s, dna, adn, from s in ?sequence,
9       dna in DNA(s,1,6), adn in DNA(s,6,1)
10 // gives
11 my_sequence atgttg caacat
12 // PEPTIDE will translate the sequence, including the UTR, in frame 1
13 select PEPTIDE (@,1,2)
14 // gives Methionine-Leucine (atg codes for Met, ttg for Leu)
15 ML
16 // CODING will translate only sequences with tag CDS, up to the stop
17 // To get all proteins encoded in messenger RNA, try
18 select ?sequence CDS // the CDS tag means the sequence is coding
19 select -o f ?sequence ; CODING // all proteins are written to file f

```

Notice that acedb will translate nuclear DNA using the standard genetic code, but if the relevant standardized information has been provided in the database, it will translate some sequences using a different genetic code, for example the human mitochondrial code to translate human mitochondrial mRNAs.

In some cases, one may have the name of a chromosome, say '12', exists in 2 different classes, say class 'Map' and class 'Sequence', or simply is a number, we can 'reclass' the variable 'm' using OBJECT before extracting the DNA:

```

1 select x,s,dna from x=11+1, s in OBJECT('Sequence',x), dna in DNA(s,101,112)
2 select m,s,dna from m in ?Map where m == '12',
3       s in OBJECT('Sequence',m), dna in DNA(s,101,112)

```

In principle, one can select all DNA sequences coding for a particular motif or coding for a particular peptide motif say MLR or MIR (Methionine (Iso)Leucine aRginine) using simple text comparison (like) of full UNIX regular expressions (equal tide)

```

1 select s,dna from s in class sequence, dna in DNA(s) where dna like '*taag*'
2 select s,p from s in class sequence, p in PEPTIDE(s) where p =~ '^M[IL]R'

```

However, the content of the sequences is not indexed, and dedicated tools like BLAT or BLAST would be much faster for large scale searches of multiple motifs.

7 Column titles and row order

The result of a query is a tab delimited table. The order of the columns is specified by the order of the variables in the *select* clause. If the option *-t* is provided, the first line, prefixed by a #, will contain the title of the columns. By default, the title of the columns are the names of the variables,

but a more complete title can be provided for some columns using the *TITLE* field (in capitals) as in:

```
1 select -t x, y, z from ... where ... TITLE x:this is x, z:this is z
```

Using the expanded options *-o file_name -title my_title*, as in

```
1 select -o foobar.txt --title 'Hello World'
2 x, y, z from ... where ... TITLE x:this is x, z:this is z
```

adds at the top of the output file 'foobar.txt' a first line prefixed by *###* with the current date, the chosen title, and the name of the output file.

The order of the rows is by default the alpha-numeric ordering. But this order can be modified using the following syntax.

```
1 // By default the lines are sorted alphanumerically
2 select x, y, z from ... where ... // default order
3
4 // The order can be modified using order_by
5 select x, y, z from ... where ... order_by 1+2+3 // default order
6 // to order by y increasing, then x decreasing, then z increasing
7 // these 2 constructions are equivalent
8 select x, y, z from ... where ... order_by 2-1+3 // use column numbers
9 select x, y, z from ... where ... order_by y-x+z // use variable names
```

In the leading column, or the following ones in case of a tie, numbers are sorted numerically : (-5 -2.7 -2.6 0 3 7 11 11.1 12) and names are sorted alphabetically. However, we found [?] that it is much nicer to order names containing embedded number numerically as in (a9 b7 b7.1a b11). The rule is to order numerically on successive groups on contiguous numbers. Notice that this method also works well for times and dates '9:20' will come before '10:2'.

8 Conclusion

As from 2018, the official site for the acedb object oriented database system has been transferred from the Sanger centre to the NCBI. We presented here the new query language developped to allow a faster and more fluid access to the data, while maintaining a good compatibility with the previous systems.

The main idea was to clearly define a full fledge syntax while at the same time developing a set of simplification rules which allow in the simple cases to write very terse queries and let the computer interpolate the syntactic sugar. This double strategy allows an integrated support for easy to read verbose scripts and for easy to type terse interactive commands. All queries are analyzed by the same machinery which interprets the query, checks the syntax, and executes the search in C while taking advantage of current state of the acedb data caches.

Some aspects of this presentation may be interesting to a broader audience not necessarily using the acedb system. For example the implicit handling of multivalued fields is much simpler and more natural than in relational databases. The ternary logic: True, False, Null presented in section 3.5 is important in any situation where the information is often incomplete. The distinction between 'A not equal B' and 'not A equal B', detailed in section 3.6, is critical whenever the variables are multivalued. The native support for DNA and proteins is handy in biological applications. Finally, the strict and loose date comparisons of section 4.2 can be appreciated as the cherry on the cake.

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A Annex

A.1 List of operators and reserved keywords

To summarize, the full list of operators, in their order of precedence is

```
1      ; TITLE order\_by
2      from select , where
3      in := =
4      || OR ^^ XOR && AND ! NOT
5      like =~ ~ == != >= <= > < >~ <~
6      ISA
7      modulo + - * / ^
8      DNA PEPTIDE PEP CODING DATEDIFF
9      COUNT MIN MAX SUM AVERAGE STDDEV
10     .class .name .timestamp
11     >> -> # =>
12     @ OBJECT
13     () {} []
```

with the restriction that the 3 kinds of parentheses and also the quotes must be nested correctly (["bb"]) is correct but (x[y]) is illegal. Operators written in capitals, i.e. DNA or AVERAGE, must be written in capitals (are case-sensitive)

A.2 History of the development of the acedb query language

The query syntax described in this document results from the convergence of several developments - the original acedb query language of 1990, available in the 'query' box of the acedb graphic interface - the table maker system of 1992, with its user friendly graphic interface - the original AQL query language developed at the Sanger around 2002

The general idea was to learn from the 3 existing systems, conserve the best aspects of each, fix their limitations and clean up the interface. For example,

- The acedb query language is very terse and expressive, but it is limited to constructing sets of objects and does not allow to display the content of selected fields.

- The table maker was meant to fix these limitations. Using its rich graphic interface, one can easily construct a tables involving objects, numbers, texts and even DNA sequences. However, even a simple tables cannot be constructed on the command line. One need always to construct the table graphically, to save its definition in a definition file, and run the query using that file.

- The original AQL query language was ment to fix this limitation. AQL defined a command line syntax, reminiscent of SQL, but adapted to the idiosyncrasies of an object oriented database like acedb. It made it in principle possible to compose a table on the command line or using a library call. Unfortunately, the syntax was slightly too rich, encouraging the user to write convoluted queries, and always verbose. Since there was no graphic interface to help compose a valid query,

and since the original AQL compiler did not explain the eventual errors, it was hard to write a non-trivial valid query. Finally, AQL execution was slower than table-maker, and lacked access to DNA and protein sequences.

From the user point of view, the previous interfaces are maintained, but the old style queries are translated internally and executed by the new query engine. In particular, the graphic table-maker interface remains valid and can still be used to construct complex queries and most original AQL queries remain valid. In practice the shortcuts defined in section 2.5 map the terse query language of the acedb graphic interface, first released in 1990, into a (modification of) the full fledged AQL queries developed around 2002. As a result, these 2 types of queries, and the table constructed via the acedb graphic table maker interface, are all expressed in the new unified syntax and the database exploration can be executed using a single highly optimized query engine.

A.3 Code availability

The acedb source code can be downloaded from

[https : //github.com/ncbi/AceView](https://github.com/ncbi/AceView), [https : //github.com/jtmieg/AceView](https://github.com/jtmieg/AceView)

The tar ball is effectively identical to the distribution of the AceView/Magic pipeline [?] which uses acedb as the underlying database, and can organize the workflow and manage the meta-data of tens of thousands of next-generation sequencing runs. The same package underlies the Gene annotation public web site <https://www.ncbi.nlm.nih.gov/AceView> [?]

The program is written in C and has very few external dependencies. To compile the code try

```
1  tcsh
2  gunzip -c magic.*.tar.gz | tar xf -
3  setenv ACEDB_MACHINE LINUX_4_OPT
4  cd magic*
5  make libs tace
6  make -k all
```

The code can also compile on the Mac using `setenv ACEDB_MACHINE MAC_X_64_OPT` or on any other linux/unix platform. Multiple choices are offered in the directory `wmake`. The `tace` command-line interface presented in the present document should compile without difficulty. The `xace` graphic-interface requires the installation of the public X11 development environment as explained in the README file situated at the top of the acedb distribution.

A test-bench containing examples of queries can be constructed using the command

```
1  tcsh wbl/bqltest.tcsh
```

After running the script, the file `ACEDB.QUERY_LANGUAGE_TEST/test.out` contains the output of the commands contained in `test.cmd` and the file `test.diff`, which contains the differences between `test.out` and `test.out.expected`, should be empty. The database can be examined and further queries can be tested using the command

```
1 bin*/tace ACEDB_QUERY_LANGUAGE_TEST
```

which will activate the command line interface of acedb. All the examples in the present document can be copied in the interface and should give proper answers. A question mark will invoke the on-line help and list all the possible commands. In particular, the acedb command 'show' will display the full content of the active objects and will help to understand why they were selected by the query.

Please try the code and send us feedback.

A.4 Running an acedb session

As explained above, a small test database can be constructed automatically. More generally, the directory waligner/metadata contains some examples of large acedb schemas. Once you have constructed a local directory containing an empty sub-directory database and a sub-directory wspecc defining the schema and a data file xxx.ace, you can initialize a database and parse an ace file as follows:

```
1 bin*/tace . // launch the compiled text-acedb on the local directory
2 y          // yes, initialize the database
3 parse myfile.ace // parse some data
4 save // save the data in compiled binary format
5 quit
```

Acedb can be regarded as a data compiler. It can read successively several data files, merge them and compile them into a compact binary format allowing fast retrieval. Clear messages signal places in the data file which do not conform to the schema. You can fix the file and reload it. Reading the same file several times is idempotent, i.e. it does not cause any problem.

Subclasses are configured in the file wspecc/subclasses.wm

```
1 Class Prolific_author // The class name must start with a letter
2 Visible // Make it available in graphic menus
3 Is_a_subclass_of Person // Inherit from class Person
4 Filter "COUNT Papers >= 5" // Chosen filter
```

One can then use the database and ask queries on the command line or import it from a file using the `-i` parameter or redirect the output to a file using `-o`

```
1 bin*/tace . // start the system
2 select ... from ... where // a query
3 xx yy zz // the reply comes on stdout
4 xx yy zz
5 select -o f.out ... from ... where ...
6 // the reply goes in the file f.out
7 select -a ... // the reply is quote protected (.ace format)
```

```

8      select -i f.bql ... // file f.bql contains the query
9      select -i f.bql -s // silent mode
10     select @ ... // use the new active set
11     ? select // help of the command line parameters
12     ? // list all the acedb commands
13     quit

```

In silent mode (with semi column at the end of the query, or using the option -s (-silent)) the output is suppressed but the active list is modified.

Acedb contains dozens of commands available by typing ? or ?command. There is also a rich graphic interface, available using the command *bin.* /xace.*, which generates on the fly in HTML5/SVG language all the plots and diagrams presented on the AceView server [?]. It provides among other tools a graphic interface helping to compose complex tables. But the description of the graphic acedb system is out of the scope of this user guide.

The MAGIC pipeline [?] is a good example of a complex system relying on acedb.

A.5 C language API

The acedb system has a C language API. The application code can either be linked into the open-source database server code, in which case it directly shares the caches of the database engine, or it can be part of a client code which calls the server via tcp. The interesting point is that exactly the same code works in both situations, one changes from a server to a client code by simply changing the library used at link stage. Here is an example of a complete C program invoking the query language

```

1  #include <ac.h> /* acedb C language API header */
2
3  int main (int argc, const char **argv)
4  {
5      AC_HANDLE h = ac_new_handle () ;
6      const char *errors = 0 ;
7      const char *dbNameS = "acedb_directory" ; // server case
8      const char *dbNameC = "t:machine_name:port_number" ; // client case
9      AC_DB db = ac_open_db (dbNameX, &errors); // X = S|C
10     const char *query = "select p in class person where p like a*" ;
11     AC_KEYSET aks = 0 ; /* initial keyset, populates the @ symbol */
12     AC_TABLE t = ac_bql_table (db, query, aks, errors, h) ;
13
14     printf ("Found % lines in the table\n", t->rows) ;
15
16     ac_db_close (db) ;
17     ac_free (h) ;
18     return 0 ;
19 } /* main */

```

The interface is detailed in the self documented header file wh/ac.h.
All comments are welcome.